

Ethics in quantitative sport management research: the impact of AI

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Abstract

Purpose – Despite the use of plagiarism-checking software and current ethical guidelines in sport management journals, raising awareness of ethical concerns and potential risks of artificial intelligence (AI) applications is necessary. This paper discusses how AI affects ethical research and publishing and provides guidelines for sport management scholars to ensure quality and integrity of their research.

Design/methodology/approach – A comprehensive review and critical analysis of literature was performed to evaluate research ethics, potential risks, and guiding principles for the use of AI in research.

Findings – Ethical research guidelines for quantitative sport management research were proposed. The guidelines encompass seven principles for the proper use of AI and ethical conduct specific to the research methods, data analysis, and results, which would be challenging for AI to accurately replicate.

Originality/value – This study provides an original contribution to the field of sport management because numerous questions concerning ethics and AI have not been addressed until now. The guidelines are suitable for use by sport management scholars, concerning the accuracy, validity, and quality of research while mitigating ethical risks in AI-generated content.

Keywords Artificial intelligence, Research ethics, Research methods, Data analysis

Paper type Research paper

Ethical behavior of researchers is paramount so that consumers of research (e.g. other researchers, practitioners, and the public) can be assured of the accuracy, validity, and quality of the research. Ethical guidelines have been established and promoted by various research and academic entities (e.g. COPE – Committee on Publication Ethics, 2024; IJSMS – International Journal of Sport Marketing and Sponsorship, 2024; Journal of University Teaching and Learning Practice – Crawford *et al.*, 2023; International Conference of Machine Learning; Nguyen *et al.*, 2023), and most academic institutions teach units or courses on ethical research (Kimmel, 2007) or have ethics review boards (Head, 2020) to educate researchers on established methods of conducting research in an ethical manner.

However, there have always been concerns about whether scholars, in general (Crawford *et al.*, 2023; Swazey *et al.*, 1993) and in sport management (Pfleegor *et al.*, 2019), are following ethical guidelines. Specifically, Crossref and Retraction Watch have identified over 50,000 article retractions in total (Hendricks *et al.*, 2023). Considering that each year over 2 million peer-reviewed research articles are published (Musa, 2023), this indicates that only a small percentage of research projects and publications every year are questioned as to their ethical



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principles. However, based on a survey of journal editors, [Smart and Gaston \(2019\)](#) estimated that around 15% of journal submissions had plagiarism issues.

Artificial Intelligence (AI) has become a point of contention in recent years due to the potential impact it may have on academic research. According to [Lund *et al.* \(2023\)](#), based on [Stanford Encyclopedia of Philosophy \(2021\)](#), AI “poses a threat to the integrity of science by introducing potential biases and errors that are difficult to identify and correct” (p. 574). [Retraction Watch \(2024\)](#) reported that since November 2021, over 90 research papers have shown evidence of at least partially being written by AI. More concerning, however, is the report that AI was able to “completely fabricate” a research article in a matter of hours with limited effort from a human user ([Májovský *et al.*, 2023](#), p. 5) and researchers have used AI to create a fake data set to “support an unverified scientific claim” ([Naddaf, 2023](#), p. 895). Furthermore, the structured, consensus-based guidelines provided by publishers and journals to protect the integrity of scientific output are limited ([Ganjavi *et al.*, 2024](#)). This has alarmed editors and reviewers alike because AI may provide more opportunities for researchers to violate ethical guidelines and may make finding such violations much more difficult to detect ([Gelinis *et al.*, 2017](#)).

Recent literature concerning ethics and AI has started to address these issues in fields other than sport management (e.g. social and behavioral sciences, environmental sciences, arts and humanities, and physical and chemical sciences; [Christou, 2023](#); [Ganjavi *et al.*, 2024](#); [Smart and Gaston, 2019](#)). Although there is no data that we know of that tracks this information across sport management journals, we suspect that a small percentage of submissions to sport management journals are rejected due to plagiarism, questionable data, irrelevant or inaccurate citations, slicing data too “thinly”, or other issues. Most sport management journals (e.g. Journal of Sport Management, Sport Management Review, and IJSMS) use plagiarism-checking software and provide ethical guidelines for authors. However, the main responsibility for ensuring the ethicality of submissions falls on the reviewers and editors and we feel that our field of sport management needs to be more aware of potential ethical issues that AI might generate.

Thus, the purpose of this paper is multifaceted. We will:

- (1) address ethical principles specific to research,
- (2) discuss how AI relates to ethical research and publishing,
- (3) provide guidelines that sport management scholars may use to reassure consumers of their research that their research:
 - follows ethical guidelines,
 - is accurate, valid, and good quality,
 - is *not* being generated by AI,
 - can be foundational for future research, and
 - can be used to provide guidance for practitioners.

Research Ethics

[Ratti and Graves \(2022\)](#) suggested that “data scientists should cultivate ethics as a form of reflection on how technical choices and ethical impacts shape one another” (p. 1819). Furthermore, since [White *et al.* \(2024\)](#) noted that there is “little clarity around . . . ethic-theoretical foundations” (p. 179) of AI and ethics and [Bonnemains *et al.* \(2018\)](#) concluded that the “ethical frameworks we have studied do not seem to be relevant in all situations” (p. 57), we have followed White *et al.*’s (2024) suggestion, and rather than trying to determine the

perfect ethical theory on which to base this treatise, we will provide “the principles upon which [our] systems are based” (p. 179). This is similar to the microethics approach proposed by [Ratti and Graves \(2022\)](#), which prioritizes the ethical relevance of technical acts by data scientists. They suggest that this approach has two components. The first component is that data scientists need to learn the ethical dimensions of a situation from role models in their particular cognate area. The second component is being able to understand the moral ability for “identifying and grasping the ethical relevance of a situation” (p. 1833). We hope to be role models for quantitative sport management researchers in this endeavor by providing a framework of ethical guidelines that both support the use of AI in some situations and discourage it in others.

[Resnik \(2020\)](#) defined ethics as “norms for conduct that distinguish between acceptable and unacceptable behavior” (para. 1). The goal here, however, is to address the more specific concept of *research ethics*: the “ethics of the planning, conducting, and reporting of research” ([Resources for Research Ethics Education, 1999–2016](#), para., 1), which includes the following areas:

- (1) Protections of human and animal subjects.
- (2) Collection, use, and interpretation of research data.
- (3) Methods for reporting and reviewing research plans or findings.
- (4) Relationships among researchers with one another.
- (5) Relationships between researchers and those that will be affected by their research.
- (6) Means for responding to misunderstandings, disputes, or misconduct; and
- (7) Options for promoting ethical conduct in research (para. 2–3).

Following their European colleagues, American institutions established the concept of the Institutional Review Board (IRB) as one mechanism through which to protect human subjects ([U.S. Department of Health and Human Services, 1979](#)). These university-based agencies, created to oversee the ethical application of social-science research, developed guidelines for the submission of written protocols for review and approval prior to the deployment of institutional-supported research ([Smith, 2016](#)). IRB principles align with WAPOR’s Code of Professional of Ethics and Practices ([WAPOR, 2021](#)). WAPOR’s standards have been both developed and disseminated in order to (a) “advance the use of science in the field of public opinion research”, (b) “protect the public from misrepresentation and exploitation in the name of research”, and (c) “maintain confidence that researchers in this field are bound by a set of sound and basic principles” (2021, para. II.A, II.C, and II.E).

[Resnik \(2020\)](#) suggests that there are five primary reasons for ethical norms when conducting research. Ethical standards should:

- (1) Focus on knowledge, truth, and avoidance of error, with no misrepresentation, fabrication, and/or falsification throughout the process.
- (2) Promote values that are essential to collaborative work including trust, accountability, mutual respect, and fairness.
- (3) Ensure that researchers can be held accountable to the public and adhere to local, state, federal, and international guidelines.
- (4) Build public support for research so that external publics are more likely to accept and fund research due to the quality and integrity of the project and scholar(s).

- (5) Promote a variety of other important moral and social values, including social responsibility, human rights, animal welfare, compliance with the law, and public health and safety (paras. 7–11).

Furthermore, [Pfleegor et al. \(2019\)](#) proposed three suggestions specifically for the sport management academe:

- (1) “Early and frequent ethical training ... for doctoral students and early career academics” (p. 204),
- (2) “Academic and organizational leaders should be focused on promoting ethical research endeavors” (p. 204), and
- (3) Academic “leaders should establish ethical scholarship policies” (p. 204).

[Resnik \(2020\)](#) also provided an overview of 18 *guiding ethical principles* to use when conducting academic research, summarized from the work of [Shamoo and Resnik \(2015\)](#). They sought to provide resources based on various policies and codes through many well-regarded agencies (e.g. National Institutes of Health, National Science Foundation, American Psychological Association, etc.). We are listing the most relevant principles to our topic here: honesty, objectivity, integrity, carefulness, openness, transparency, accountability, intellectual property, and human subject protection ([Resnick, 2020](#)). For academic disciplines who do not have such support, this is a solid foundation from which to adopt and/or adapt. Next, we examine the potential impacts of AI (both positive and negative) on this endeavor.

Artificial Intelligence

John McCarthy defined AI in 1956 as “the science and engineering of making intelligent machines” ([Collins et al., 2021](#)). The advancements and increased utilization of AI in recent years have given a more sophisticated understanding, describing AI as technologies that mimic human intelligence to perform tasks using software and algorithms ([Christou, 2023](#)). Current AI systems have become more capable of processing and analyzing massive amounts of data, identifying patterns and trends, generating insights, and automating complex tasks ([Langley, 2012](#)).

AI comprises various applications such as machine learning and deep learning. Machine learning includes techniques that use computer algorithms and analytics to refine patterns to make more accurate classifications or predictions over time ([Collins et al., 2021](#)). For example, YouTube and Google use machine learning algorithms to provide content recommendations by understanding users’ preferences ([Dejoux and Léon, 2018](#)). Deep learning is a subset of machine learning, based on deep neural networks consisting of multiple input, output, and hidden layers. These layers process lots of data to simulate the complex decision-making power of the human brain ([Christou, 2023](#)). Deep learning can be used as a major force for automatic speech recognition, image recognition, natural language processing, and bioinformatics ([Cao et al., 2018](#)). GPTs (Generative Pre-trained Transformers) are types of deep learning models widely used by the qualitative research academic community for natural language processing (NLP) tasks such as translation, text summarization, and information analysis ([Christou, 2023](#)).

Beneficial use of AI in academic research

The ability of AI to combine and apply knowledge and skills based on diverse sources and learning has brought numerous benefits to academic writing. Various AI-powered tools, such as Grammarly, ChatGPT, Research Rabbit, ChatPDF, Semantic Scholar, Iris.ai, and SciSpace,

offer services and support to researchers in several tasks (Crawford *et al.*, 2023; Khalifa and Albadawy, 2024; Morgan, 2024). First, AI tools can help synthesize relevant information from a wide range of scientific papers quickly and efficiently. They assert their capabilities to search, read, and summarize lengthy research literature (Lee *et al.*, 2022). The synthesis of literature leads to a comprehensive and nuanced understanding of existing research, enabling researchers to swiftly understand the key points of the paper and draw out meaningful insights (Ciaccio *et al.*, 2023; Dergaa *et al.*, 2023).

Second, AI offers transformative solutions for analyzing, managing, and interpreting data. Algorithms can process, analyze, and interpret immense datasets, which identify complex patterns, correlations, and trends (Wohlfarth *et al.*, 2023). Detailed analysis and visualization often provide valuable insights that might not be rapidly apparent through traditional methods (Khalifa and Albadawy, 2024). Such data-driven insights help researchers not only generate hypotheses or research questions but also assist in their validation (Pion-Tonachini *et al.*, 2021).

Third, AI tools are useful for proofreading and editing so that researchers can enhance the textual quality of their manuscripts (Khalifa and Albadawy, 2024). Beyond spell-checking, AI tools help the researcher reword and rephrase sentences or paragraphs (Sasaki, 2023). These tools can guide appropriate language usage and make the manuscript clearer and more impactful (Ciaccio, 2023). This enhancement is crucial for effectively communicating complex scientific concepts and ideas (Ciaccio, 2023). Continuing to leverage the effectiveness of AI tools, researchers can input the relevant details of their sources to format their reference lists (Zhang, 2012). The AI-powered citation generators also support various citation styles such as APA, MLA, Chicago, and Harvard styles. Therefore, the AI academic writing tools enable researchers to accelerate and improve their overall writing process, allowing them to dedicate more time to critical thinking, hypothesis generation, and exploring new research avenues (Khedkar, 2023).

Lastly, AI survey question generators may provide benefits by helping researchers easily brainstorm and develop relevant and effective survey questions (Balayn *et al.*, 2021). This AI survey tool can create automated questions in response to the survey's purpose and topic (Esmaeilzadeh, 2020). The questions can also be tailored to individual respondents based on their demographic characteristics and prior responses (Guo and Yu, 2022). This simplified survey generation process could potentially help researchers create reliable questions, which might enhance the overall data quality.

Misuse of AI in academic research

While the benefits of AI in academic research are considerable, the use of AI can present multifaceted challenges that demand careful consideration. As researchers delve into the realm of AI integration, it is imperative to remain vigilant about the potential ramifications and ethical implications inherent in its utilization (Zhu *et al.*, 2022).

First, AI-powered tools and the work generated with AI may be based on biased or harmful content. Such biases are often caused by issues in the training dataset or the decision algorithms (Ferrer *et al.*, 2021). Several studies have revealed that AI-based tools, often trained on web-based datasets, can contain biases concerning gender, race, ethnicity, and disability status (Zhao *et al.*, 2019). Such biases can influence the formulation of survey items with skewed perspectives. Consequently, these biases can perpetuate unfair treatment and systemic discrimination against certain social groups, posing significant ethical concerns (Bolukbasi *et al.*, 2016; Buggineni, 2023; Christou, 2023).

Second, the proliferation of misinformation and lack of transparency surrounding AI-generated content exacerbates the ethical challenges. AI-powered tools, particularly those employing deep learning models, may access and process sensitive personal information,

raising privacy issues (Nguyen *et al.*, 2023; Schwartz *et al.*, 2020). Furthermore, the opacity surrounding the sources and mechanisms behind AI-generated content engenders skepticism about its reliability. For example, incomplete or missing data creates uncertainty about how AI algorithms will process it. Moreover, normalizing data for machine learning models can impact its quality, potentially affecting the model's performance (Buggineni, 2023). This lack of transparency can facilitate the creation and dissemination of plausible fake information, fueling the spread of disinformation and propaganda (Lee *et al.*, 2022; Qiu, 2023). Consequently, the reliance on AI-generated content for scholarly research may compromise the integrity and rigor of academic endeavors.

Third, the advent of AI-powered tools challenges established paradigms of authorship and intellectual property. Content generated through AI algorithms blurs the lines of originality, making it arduous to discern the true origin of ideas or concepts (Dogru *et al.*, 2023; Lund and Naheem, 2024). This ambiguity surrounding authorship raises pertinent questions about academic integrity and the fair attribution of intellectual contributions.

Fourth, it is also possible to create "synthetic" data based on real data or generate data from scratch if one has the specific parameters one wants the data to meet (Synthetic Data Generation, 2024). Although, synthetic data generation might offer a valuable solution in some cases, as it creates artificial datasets for AI model training and testing (Buggineni, 2023), it can also make it very easy for researchers create realistic but fake data sets, such as non-existent participants data, fake answers to survey questionnaires, or generate a large data set. Such practices might evade detection by plagiarism software, raising serious ethical concerns (Naddaf, 2023).

Ethical guidelines for quantitative sport management research

Based on Ratti and Graves (2022) microethics approach, Pfleegor *et al.*'s (2019) suggestions about providing ethical training and promoting and establishing ethical research endeavors, and Resnik's (2020) ethical principles, in addition, to the potential for misuse of AI in quantitative sport management research, we propose a list of ethical research guidelines for the discipline. These standards are specific to the data science process (Ratti and Graves, 2022), which includes data acquisition, data understanding and preparation, data analysis and modeling, validation and interpretation of the model, and communication of results (p. 1835). This data science process would be difficult for AI to replicate, at least currently, due to the degree of difficulty for AI to obtain the relevant information in the quantity that would allow it to create a believable Methods and Analysis/Results section. We are not saying that it cannot be done by AI, but AI would have to have, not only the survey items and the data set itself, but access to a sufficient number of relevant published sport management articles to learn how to accomplish all of the processes below. Although more and more quantitative sport management articles are available via open access, most of the quality articles that address all of the information below are still in research databases (e.g. Academic Search Complete, JSTOR, ProQuest Research Library, etc.) that are by subscription only or must be bought or rented online, which AI does not presently have access to (Butler, 2023). We feel that following these guidelines will produce a Method and an Analysis/Results section that will reassure the consumers of the research that the research is ethical, accurate, valid, of good quality, and not generated by AI. Thus, the results can be foundational for future research and provide guidance for practitioners.

Data acquisition (item generation)

In this section we focus on the research methods that quantitative researchers should follow to reduce issues that could be both ethical problems and research quality problems.

Item choice or generation. When creating or compiling a survey there are several things that researchers need to consider a) content validity of the items, b) modification of existing scales, c) inclusion of multiple components in items, and d) compound items. There is software that uses “AI and machine learning to make surveys better” (QuestionPro, 2024), however nowhere in its literature does it address any of the concerns noted below.

Content validity. Content validity refers to the idea that the item measures what it is supposed to measure (Bollen, 1989) and the items should be developed from the appropriate theoretical framework (Trail and James, 2016). For example, in measuring team identification based on identity theory, an item such as “I am highly identified with the team” has good content validity as would “I am a fan of the team”. Trail and James (2016) provide a good summary of content validity, but the key issue here is the use of scales that do not have good content validity. If a researcher knowingly uses scales lacking content validity, then that is unethical, and poor research as well. Even if there hasn’t been direct criticism of the content validity of a scale, researchers need to assess such on their own, and discuss as needed. All of the items used in the survey should be presented in a table, so that reviewers and consumers of the paper (if published) can also assess content validity. This gets to the ethical principles of honesty, integrity and carefulness.

Modification of existing scales. When researchers modify existing scales, they must provide a rationale for doing so, such as modifying to improve content validity. However, as reviewers, we have seen researchers modify items and scales and provide no rationale at all. This might just be poor research, but it could also present ethical issues if modifications aim to increase common variance in an artificial manner (see below for a more detailed discussion). Therefore, following the ethical principles identified above of openness and transparency, it is critical that any modification be identified, and a viable rationale provided.

Inclusion of multiple components in one item. Items should not have multiple components (e.g. cognitions, affect, and/or behaviors) all in one item, which would be incorrectly operationalizing the concept (Jones, 2015). For example, the item “I am a fan of the team” is a cognitive representation of team identification. It only has one component (a cognitive assessment of level of fandom). However, if the item was “I wear team paraphernalia to show that I’m a fan of the team”, that item has both behavioral (wearing behavior) and cognitive components (fan of the team). The item has other issues as we will discuss below, but when an item has multiple components, then it is difficult to assess whether it is a cognitive, affective, or behavioral item. Considering that many argue that past behaviors predict current cognitions, or current cognitions predict future behaviors (Trail, 2019), making/using items that are as simple and clear as possible can eliminate any analytic issues later on. Intentionally creating/using items that might increase the variance explained in a DV by an IV by overlapping components is unethical (see discussion of Common Method Bias below) and should not be done because it violates the integrity and objectivity ethical principles.

Compound items. Items that include conjunctions (e.g. because, since, as, etc.) should not be used because they create subordinate clauses that are purported to be the reason for the initial clause. For example, in Mahoney *et al.*’s (2000) PCT scale, an item is “I have been a fan of the team SINCE I began watching collegiate football”. The conjunction “since” provides a reason for being a fan. Items that provide reasons through conjunctions assume that these reasons are why someone is or does something. This should be avoided. Instead, two items should be used, one that is the preface phrase (I am a fan) and then a separate item that measures the watching behavior (the subordinate clause). Then an analytic procedure such as a correlation can be used to look at the relationship between the two, rather than forcing the respondent to answer two questions within one item. This reflects the ethical principle of carefulness, especially when done knowingly. Using AI to create survey items, or surveys, is not unethical on the face of it, as long as the process is described in full, and all the above concerns are addressed.

Ethical Guideline #1. Survey items should have content validity, have single components, and avoid compound phrases. If items are modified from existing scales, then the rationale for any changes must be noted. If AI is used at any point in the item generation or modification process it must be noted. If AI uses existing items to “create” its own items, this must be determined, and the original source of the item cited.

Data acquisition (survey organization)

Appropriately organized surveys/questionnaires can reduce a) common method bias issues, b) artificially inflated commonality due to identical terms across items, and c) elevated correlations between items due to location of the items on the questionnaire.

Common method bias. Consumers of research should be concerned about potential common method bias (CMB) or common method variance (CMV) in the data. All researchers should address this issue if relevant. CMV is the “variance that is attributable to the measurement method rather than to the constructs the measures represent” (Podsakoff *et al.*, 2003, p. 879), which may create a bias in favor of the researcher’s results.

Artificially inflated commonality. Increasing variance explained in first order latent constructs by intentionally causing CMB is unethical. For example, listing all items in a multi-item scale sequentially on a questionnaire artificially increases the common variance among the items because many respondents may identify the similarity among the items and answer all of them alike, a type of order bias (Aaker *et al.*, 2004). If this is done intentionally with the knowledge that it will increase Cronbach’s alpha values and AVE (Average Variance Explained) values, it is unethical and violates the honesty and transparency principles listed above. If this is done unintentionally due to a lack of knowledge, then it is poor research (Trail and James, 2016).

Elevated correlations. One way of increasing CMV between IVs and DVs is by including terms in items that artificially inflate commonality. For example, if the IV item is “I am a fan of the Boston Celtics” from the Team Identification Index (TII; Trail and James, 2001), and the DV is “I intend to watch Boston Celtics games on TV” from the Intentions for Sport Consumption Behavior Scale (ISCBS, Trail *et al.*, 2003), then including the words “Boston Celtics” in both items will increase the common variance between the two items solely due to the wording of the items, especially if the items are near each other in the questionnaire (Aaker *et al.*, 2004). Intentionally doing this, with foreknowledge of the potential for elevating common variance, is certainly unethical and violates the principles of objectivity and openness.

Ethical Guideline #2. Questionnaires should be created to reduce CMB by either randomizing the items or intentionally distancing items that are within the same multi-item scale and by distancing IV items from DV items. Additionally, items should not contain the same words which artificially inflate the correlations among items. Using AI to check for these issues would be ethical. However, using AI to artificially increase common variance would be unethical.

Data acquisition (collecting the sample)

In general, most researchers are well-educated on how to collect a sample. The sample should be representative of the population of interest as much as possible and can be achieved by randomly sampling the population of interest. However, that is not always possible due to cost, access, or other constraints. Regardless, the researcher must accurately describe how the sample was collected, its characteristics, and the population it represents.

Ethical Guideline #3. Researchers should describe the population, attempt random sampling, and if not possible, explain why not. Detailed descriptions of the sample collection

process reassure readers of its authenticity, especially in light of AI's ability to generate synthetic data.

Data preparation (cleaning the data)

Once the data is collected, the researcher needs to assess the data set to determine if the responses are viable and should be included in an analysis. Three aspects to consider when "cleaning" the data are: a) time spent filling out the survey by the respondent, b) Incomplete data, and c) pattern response violations.

Time on survey. Ideally respondents will take enough time to carefully read and understand each survey question, providing accurate responses. However, in reality, most respondents rush through surveys or take extended breaks, leading to potential accuracy issues. In the former, if the respondent does not read the instructions or the items sufficiently well, it is highly likely the responses may not be reflective of that respondent's reality. In the latter situation, a lengthy response time may indicate a lack of focus on the respondent's part where the respondent is doing other things while "taking" the survey. Again, the accuracy of the responses could be questioned if the respondent is not putting their full effort into answering the questions accurately.

Incomplete data. Many respondents skip questions if they are allowed to, or some just stop taking the survey at any point, for many reasons. When the researcher is forced to deal with incomplete data from a respondent, the researcher needs to know how to handle that missing data. It has been suggested that any missing data should disqualify that respondent, and that line of data should be removed from the data set (Malhotra, 1999). Although that is one option, sometimes respondents may fill out all of the survey except for the last section, which may include non-critical items to some analyses. We would encourage researchers to assess whether those respondents should be eliminated, or perhaps they could be included in some analyses, but not others. In other cases, maybe only one or two items were skipped. There are several recommendations on how to deal with missing data in these situations, especially if the missing response is only one item in a multi-item scale (Malhotra, 1999), so that the respondent does not need to be eliminated. However, if many items are skipped, those respondents must be eliminated from the data set.

Pattern response violations. When respondents repeatedly indicate the same number (e.g. 7, 7, 7, 7, 7) or follow a distinct pattern in their answers (e.g. 1, 2, 3, 4, 5), these are called pattern response violations (Malhotra, 1999). To prevent this and ensure accuracy, researchers should include negatively worded or contradictory items in each section. Although exceptions may exist, these pattern responses are often inaccurate and should likely be removed from the data. Intentionally including respondents who have pattern response violations, or who took too little or too much time to complete the survey, just for the purpose of increasing sample size is unethical and violating the principles of honesty, objectivity, and carefulness because it misrepresents the data and can lead to invalid statistical analyses.

Ethical Guideline #4. Data cleaning must remove all respondents with incomplete data or pattern response violations. The researcher needs to detail the number of respondents eliminated for each of the reasons above. If missing data is replaced using an accepted procedure, then this needs to be described as well, and the number of missing data noted, along with the number of respondents impacted. While using AI to replace missing data under one of the permissible techniques noted above is ethical, creating synthetic or fake data is not.

Data analysis

Once a "clean" data set is obtained, the researcher should address several aspects when analyzing the data: a) normality of the data, b) multicollinearity issues, c) choice of analytic technique, d) measurement model (if applicable), and e) structural model (if applicable).

Normality of the data. The normality of the data must be addressed briefly. Noting the skewness and kurtosis ranges and addressing any violations should be sufficient. Outliers should be eliminated if they skew the data too much. Not reporting violations in the normality of the data when those values changed the results favorably would violate the ethical principles of honesty, objectivity and openness.

Multicollinearity issues. Both the items and constructs (if applicable) should be assessed for violations of multicollinearity. Items that have too high correlations with each other and/or load very highly on the same construct are usually redundant and one or more items should be removed (Aaker *et al.*, 2004). Constructs that are too highly correlated lack discriminant validity, which means that they should not correlate with measures from which they should differ (Campbell, 1960). This could cause statistical discrepancies rather than real structural hypothesized relationships (Farrell, 2010), leading to erroneous interpretations of the data by the researcher. Most researchers use either the Fornell and Larker (1981) criterion or cross-loadings or both (Henseler *et al.*, 2015). However, Henseler *et al.* (2015) show that the heterotrait-monotrait ratio of correlations (HTMT) is a better technique. Using the former methods which are considerably less strict, and not reporting the HTMT with foreknowledge that the HTMT shows multicollinearity issues, would violate the ethical principles of integrity, honesty, and transparency.

Analytic technique. Choosing the right analytic technique is crucial for data evaluation. While we won't cover all possibilities here, we, as reviewers, have noticed an increase in the use of PLS (partial least squares) analysis recently. While appropriate for many relationship data analyses, researchers sometimes opt for PLS-SEM when CB-SEM (covariance-based structural equation modeling) would be more suitable (see Trail *et al.*, 2024, for a comparison of the two across several situations). Using incorrect analytic techniques may cause the researcher to misinterpret the data. In some instances, using one technique over another can allow the data analysis to actually work, when if the correct technique were used it would show that the model was not empirically sound. Obviously, doing this intentionally would be unethical.

Measurement model (correlating error terms). Another issue that occurs too frequently is correlating error terms in measurement or structural models. Correlating the error terms when not appropriate will potentially improve the model fit improperly (MacCallum *et al.*, 1992). Doing this with foreknowledge that it is inappropriate is unethical and doing it unknowingly is poor analysis. Papers have been rejected due to this.

Structural models. Lately, we've noticed papers where authors add paths to their structural models that weren't hypothesized or discussed in the literature review. Adding paths solely to improve model fit is unethical. Randomly adding paths to the model due to the modification indices determined in prior iterations of the model should be avoided. If the path was essential, it should have been theoretically justified beforehand, not added afterward. Another issue we've seen is where a *past* DV is "predicted" by *current* cognitive or affective measures. For example, if the DV is past attendance behavior, it is not possible to predict it by using a current measure of fan identification. The present cannot predict the past. Although this may not be an intentional ethical violation, it is certainly inappropriate.

Significance vs. meaningfulness. Statistical significance occurs when a result is unlikely due to chance at a low designated level (e.g. $p < 0.05$), suggesting an experimental effect or a hypothesized causal relationship. However, achieving significance is easier with larger sample sizes. For instance, a path coefficient could be significant at 0.1 with $N = 500$, explaining only 1% of the variance in the DV, which may not be meaningful (Cohen, 1988). Meaningfulness refers to a substantial shared variance between variables, indicating a worthwhile difference. In the above example, if fan identification explains only 1% of future attendance behavior, it's not practically meaningful for marketers. Thus, representing significant results as meaningful, and basing recommendations on them, despite knowing

they lack value to practitioners, would violate the ethical principles of transparency, honesty, and accountability.

Ethical Guideline #5. Data analysis should assess the normality of the data, look for multicollinearity issues, choose the appropriate analytic technique, assess both measurement and structural models (if applicable) correctly, and report both meaningfulness values and significance levels. Using AI to assist in any aspect of the analysis as one would use any other statistical software is perfectly ethical, as noted above; however, using AI after models are analyzed to determine how to fix the models post-hoc is inappropriate and unethical.

Validation and interpretation of the model (intentional misinterpretation of the data)

Unfortunately, as reviewers, we have seen authors misinterpret their results. It may not be intentional, but if so, it would certainly violate many of the ethical principles listed above, especially integrity, transparency, honesty and objectivity. For example, building on the above content relative to significance and meaningfulness, if researchers found a significant but not meaningful result, and intentionally tried to promote the significance as important, when it is not, then that is unethical. Specifically, if researchers found that an increase of \$100,000 in a particular marketing campaign by a team *significantly* increased future revenue per game by the 'Locals' market segment, but the results were not meaningful (less than a \$3 increase on a normal expenditure of \$67 per game per person for a 2200-person segment), then recommending that marketers continue to pour money into that campaign would be unethical. In this example it would take more than 15 home games to break even in a 17-home game season.

Ethical Guideline #6. Researchers should ethically interpret their results, without misrepresenting their findings, especially if original hypotheses are not supported. Using AI to write up and interpret the results is unethical.

Communication (general ethical issues and the use of AI when writing a research article)

Researchers should always cite the original sources of prior ideas, theories, and research. Additionally, researchers should not plagiarize, nor self-plagiarize. Furthermore, researchers should only include citations that support their information and hypotheses, and not include random cites that are irrelevant (Mehregan, 2022). Although AI can be used to find references and journal articles that are relevant to the authors' content, relying on AI can be risky because, currently, AI is not always accurate when providing citation sources (Májovský *et al.*, 2023). Furthermore, following the ethical principle of accountability, all authors should always check the data and analyses on any paper they are involved with (Resnik, 2020). If AI is used inappropriately in a paper, all authors are accountable.

Ethical Guideline #7. When participating in research of any kind, individuals should follow general ethical guidelines (e.g. proper citations, lack of plagiarism, appropriate carefulness and accountability, etc.) when handling both their own work and including the work of others.

Conclusion

In sum, we discussed ethical principles related to academic research and how AI might have beneficial or deleterious effects on that research. More specifically though, we proposed seven ethical guidelines for sport management researchers that, if followed, would reassure consumers of their research that it was accurate, valid, of good quality, and not generated by AI.

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